LAB 7: Fuzzing for Input Validation Bugs

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01_Original_Processor_Output

```
hemamohan@Hemas-MacBook-Air Lab 7 % python3 processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 75,98,37
Summary: Vitals: HR=75.0, SpO2=98.0% - STABLE

==== Running compute_patient_bmi ====
Enter weight/height (e.g. 70/1.75): 70/1.75
BMI: 22.86

==== Running assign_ward_bed ====
Enter allocation (e.g. ICU-05): ICU-05
Assignment: Intensive Care Unit Bed #5
```

02_Failing_Hypothesis_Tests p1

```
hemamohan@Hemas-MacBook-Air Lab 7 % python3 -m pytest test_processor.py
============================= test session starts ==============================
platform darwin -- Python 3.14.0, pytest-9.1.1, pluggy-1.6.0
rootdir: /Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7
plugins: hypothesis-6.170.15, anyio-4.14.2
collected 3 items

test_processor.py FFF                                                    [100%]

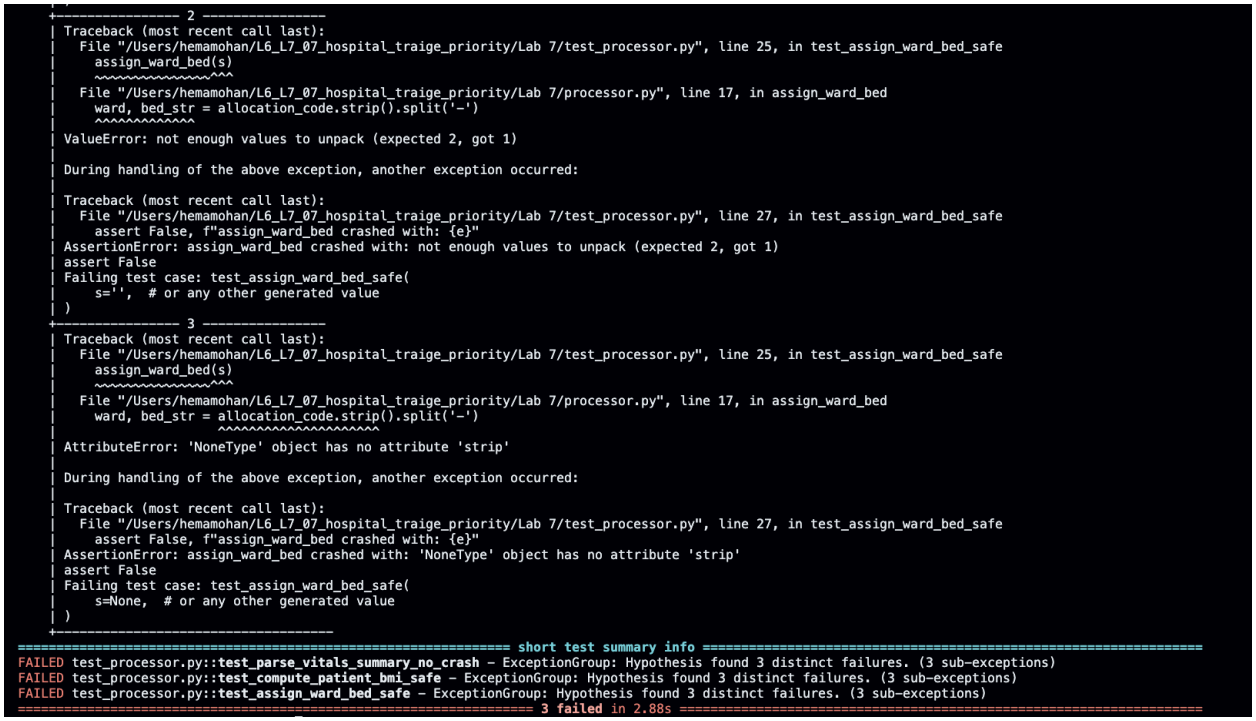
=================================== FAILURES ===================================
_____________________ test_parse_vitals_summary_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 7, in test_parse_vitals_summary_no_crash
  |     def test_parse_vitals_summary_no_crash(s):
  |                ^^^
  |   File "/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/hypothesis/core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 9, in test_parse_vitals_summary_no_crash
    |     parse_vitals_summary(s)
    |     ~~~~~~~~~~~~~~~~~~~~^^^
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 4, in parse_vitals_summary
    |     spo2 = float(parts[1])
    |                  ~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 11, in test_parse_vitals_summary_no_crash
    |     assert False, f"parse_vitals_summary crashed with: {e}"
    | AssertionError: parse_vitals_summary crashed with: list index out of range
    | assert False
    | Failing test case: test_parse_vitals_summary_no_crash(
    |     s='0',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 9, in test_parse_vitals_summary_no_crash
    |     parse_vitals_summary(s)
    |     ~~~~~~~~~~~~~~~~~~~~^^^
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 3, in parse_vitals_summary
    |     hr = float(parts[0])
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 11, in test_parse_vitals_summary_no_crash
    |     assert False, f"parse_vitals_summary crashed with: {e}"
    | AssertionError: parse_vitals_summary crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_parse_vitals_summary_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 9, in test_parse_vitals_summary_no_crash
    |     parse_vitals_summary(s)
    |     ~~~~~~~~~~~~~~~~~~~~^^^
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 2, in parse_vitals_summary
    |     parts = vitals_csv.split(',')
    |             ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 11, in test_parse_vitals_summary_no_crash
    |     assert False, f"parse_vitals_summary crashed with: {e}"
    | AssertionError: parse_vitals_summary crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_vitals_summary_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
                                                  test_compute_patient_bmi_safe
```

02_Failing_Hypothesis_Tests p2

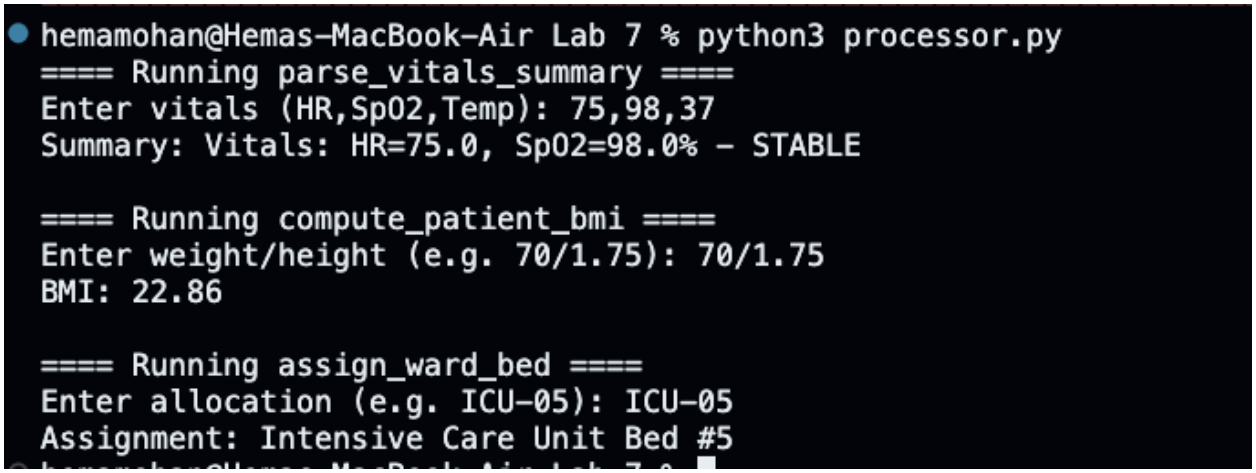


```
+------------------------------------
_____ test_compute_patient_bmi_safe _____
  + Exception Group Traceback (most recent call last):
  |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 15, in test_compute_patient_bmi_safe
  |     def test_compute_patient_bmi_safe(s):
  |                ^^^
  |   File "/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/hypothesis/core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 17, in test_compute_patient_bmi_safe
    |     compute_patient_bmi(s)
    |     ~~~~~~~~~~~~~~~~~~~^^^
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 11, in compute_patient_bmi
    |     weight = float(w_str)
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 19, in test_compute_patient_bmi_safe
    |     assert False, f"compute_patient_bmi crashed with: {e}"
    | AssertionError: compute_patient_bmi crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_compute_patient_bmi_safe(
    |     s='/',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 17, in test_compute_patient_bmi_safe
    |     compute_patient_bmi(s)
    |     ~~~~~~~~~~~~~~~~~~~^^^
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 10, in compute_patient_bmi
    |     w_str, h_str = bmi_string.split('/')
    |     ^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 19, in test_compute_patient_bmi_safe
    |     assert False, f"compute_patient_bmi crashed with: {e}"
    | AssertionError: compute_patient_bmi crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_compute_patient_bmi_safe(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 17, in test_compute_patient_bmi_safe
    |     compute_patient_bmi(s)
    |     ~~~~~~~~~~~~~~~~~~~^^^
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 10, in compute_patient_bmi
    |     w_str, h_str = bmi_string.split('/')
    |                    ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 19, in test_compute_patient_bmi_safe
    |     assert False, f"compute_patient_bmi crashed with: {e}"
    | AssertionError: compute_patient_bmi crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_compute_patient_bmi_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____ test_assign_ward_bed_safe _____
  + Exception Group Traceback (most recent call last):
  |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 23, in test_assign_ward_bed_safe
  |     def test_assign_ward_bed_safe(s):
  |                ^^^
  |   File "/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/hypothesis/core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 25, in test_assign_ward_bed_safe
    |     assign_ward_bed(s)
    |     ~~~~~~~~~~~~~~~~^^^
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/processor.py", line 24, in assign_ward_bed
    |     bed_num = int(bed_str)
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/hemamohan/L6_L7_07_hospital_traige_priority/Lab 7/test_processor.py", line 27, in test_assign_ward_bed_safe
    |     assert False, f"assign_ward_bed crashed with: {e}"
    | AssertionError: assign_ward_bed crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_assign_ward_bed_safe(
    |     s='-',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
```

02_Failing_Hypothesis_Tests p3



03_Fixed_Processor_Output



04_Passing_Hypothesis_Tests

